REINFORCEMENT LEARNING ACTIVITY

Exercise: Framing a Game as a Reinforcement Learning Problem

General Instruction

Choose one simple game or real-world problem and convert it into a Reinforcement Learning (RL) setup. Identify the problem/game, environment, agent, actions, and reward function. Then answer the Self-Play question based on your chosen game.

Part A – RL Problem Setup

Instructions

1. **Choose a game or problem.**
    - Examples: Tic-Tac-Toe, Chess, Maze Game, Rock-Paper-Scissors, Robot Navigation, or Pac-Man.
2. **Identify the Environment.**
    - Describe the world or situation where the agent operates.
    - Example: A Tic-Tac-Toe board containing X's and O's.
3. **Identify the Agent.**
    - Identify who or what makes decisions.
    - Example: The AI player.
4. **Identify the Actions.**
    - List the possible actions the agent can perform.
    - Example: Place an X in any available square.
5. **Define the Reward Function.**

- Assign rewards or penalties based on the result of an action.

- Example:
- Win = **+1**
- Draw = **0**
- Lose = **-1**

Part B – Exercise 1.1: Self-Play

Instructions

Suppose the RL agent **does not play against a random opponent**. Instead, the agent plays **against another copy of itself**.

Answer the following questions:

1. What do you think would happen when the agent plays against itself?
2. Would the agent learn a different way of playing?
3. How could self-play affect the agent's strategy?
4. What are the possible advantages of self-play?
5. What problems or limitations might occur during self-play?